

An Algebraic Convolution Formulation for Multiple-Scattering Correction in Small-Angle Neutron Scattering

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Multiple scattering in small-angle neutron scattering (SANS) redistributes spectral weight and distorts structural interpretation, particularly for thick or strongly scattering samples. We develop a finite-dimensional spectral desmearing framework that corrects multiple scattering without resorting to integral transforms or model-dependent extrapolation. The primary intensity is expanded in an orthonormal basis adapted to the isotropic transverse-momentum measure, under which convolution reduces to a recursive tensor contraction, allowing the Poisson-weighted multiple-scattering series to be evaluated directly in a finite-dimensional basis representation. This formulation yields a stable forward-inverse mapping between apparent and primary spectra. Numerical tests demonstrate convergence under repeated convolution and accurate recovery of the single-scattering intensity. Application to SANS measurements collected at multiple neutron facilities, including the Spallation Neutron Source (SNS), the High Flux Isotope Reactor (HFIR), and the Institut Laue-Langevin (ILL), shows quantitative reconstruction of the underlying primary spectrum across a wide range of transmission conditions, including strongly attenuating samples. The method provides a stable, model-agnostic framework for multiple-scattering correction in SANS and enables consistent structural interpretation across instruments and scattering regimes.

I. INTRODUCTION

Small-angle neutron scattering (SANS) is a powerful technique for probing mesoscale structure in soft matter, polymers, and complex fluids. Under the standard single-scattering approximation, the measured scattering intensity $I(Q)$ is directly related to the Fourier transform of spatial correlations in the sample¹, enabling quantitative structural interpretation across a wide range of length scales^{2,3}. In practice, however, this approximation breaks down for sufficiently thick samples, strong scatterers, or long neutron wavelengths, where neutrons may undergo multiple scattering events before exiting the sample. The resulting multiple scattering redistributes intensity in momentum space, distorting both the magnitude and the shape of the measured scattering profile and complicating subsequent analysis.

The correction of multiple scattering effects in SANS has therefore been the subject of sustained theoretical and methodological effort. Classical treatments describe multiple scattering as a sequence of statistically independent scattering events, leading to a convolution structure in momentum space⁴. Within this framework, the apparent scattering intensity may be expressed as a superposition of self-

convolutions of the single-scattering intensity, weighted by the probability of experiencing a given number of scattering events. Several practical correction schemes have been developed based on this picture, including Fourier- or Hankel-transform-based deconvolution methods that exploit the simplification of convolutions into products in a conjugate real-space representation⁵⁻⁷.

A common feature of these approaches is the explicit transformation of the measured intensity $I(Q)$ into real space, typically via a two-dimensional Fourier transform or, for isotropic data, a zeroth-order Hankel transform. While mathematically well founded, this step introduces a fundamental practical difficulty: experimental SANS data are necessarily measured over a finite and often limited range of momentum transfer Q . Truncation of the scattering profile in reciprocal space implies that its real-space representation is effectively multiplied by a sharp window function, leading to oscillatory artifacts upon transformation. These truncation-induced oscillations, closely related to the Gibbs phenomenon, can significantly contaminate the real-space signal^{8,9}, and when propagated back through an inverse transform, reintroduce systematic distortions into the corrected scattering intensity.

To mitigate these artifacts, existing desmearing algorithms typically rely on extrapolations of the scattering intensity be-

yond the experimentally accessible Q range. In many implementations, this is achieved by assuming an analytical model for the underlying single-scattering intensity that is defined from $Q = 0$ to $Q = \infty$ ^{10,11}. The model is fitted to the measured data over the available range and then used to supply the missing low- Q and high- Q behavior required for stable Fourier or Hankel transforms. While effective in favorable cases, this strategy intrinsically ties the validity of the multiple-scattering correction to the correctness of the assumed scattering model.

In practice, however, an exact or even reliable analytical model for the experimentally probed system is often unavailable. This may arise from structural heterogeneity, polydispersity, anisotropy, or the presence of competing scattering contributions that obscure the true single-scattering form. In many soft-matter and complex-fluid systems, the scattering intensity reflects a combination of emergent correlations that cannot be captured by a single closed-form expression. As a result, model-dependent extrapolations introduce an additional, often uncontrolled, source of systematic error, and the applicability of existing multiple-scattering correction schemes becomes severely limited precisely in the regimes where multiple scattering is most pronounced.

These considerations motivate the development of algebraic, basis-projected convolution frameworks for multiple-scattering correction that avoid explicit reliance on real-space transforms or on detailed *a priori* models of the scattering function, while remaining consistent with the underlying statistical structure of multiple scattering in SANS.

II. METHODS

To quantitatively account for the influence of multiple scattering on the measured scattering signal, we adopt a probabilistic description in which successive scattering events experienced by a neutron traversing the sample are treated as independent random processes occurring in a homogeneous and isotropic medium. Each elementary scattering event is governed by the same statistical law defined on the transverse momentum-transfer space. Under the assumption of isotropy, this law is invariant under planar rotations. Consequently, multiple scattering may be described as the repeated application of a single elementary scattering law via convolution.

Within this framework, the total number of scattering events occurring along a neutron trajectory is modeled as a Poisson random variable, which is a standard approximation for moderately scattering media. Let n denote the number of scattering events and μ the corresponding mean event count. The probability of observing exactly n scattering events is given by

$$P_n = e^{-\mu} \frac{\mu^n}{n!}. \quad (1)$$

The sample transmission T is defined as the probability of zero scattering events,

$$T = P_0 = e^{-\mu}, \quad (2)$$

which implies the relation $\mu = -\ln T$.

Here, T and μ refer specifically to the attenuation associated with small-angle scattering events that contribute to angular redistribution within the present multiple-scattering framework. In practice, the experimentally measured transmission may also contain contributions from absorption, incoherent scattering, and inelastic scattering, and therefore does not in general coincide with the pure small-angle scattering transmission assumed here. A more rigorous determination of the small-angle scattering attenuation would require a dedicated treatment, for example by estimating the small-angle contribution through integration of the desmeared intensity.¹²

We introduce the primary scattering intensity $\hat{I}(Q)$, defined as the intensity distribution associated with a *single* scattering event and normalized independently of transmission effects. Under isotropy, $\hat{I}(\mathbf{Q}) = \hat{I}(Q)$, and we normalize it with respect to the invariant measure of the transverse momentum space,

$$\int_0^\infty 2\pi Q \hat{I}(Q) dQ = I_0, \quad (3)$$

where I_0 sets the overall intensity scale. With this normalization, the reduced profile $\hat{I}(Q)/I_0$ may be interpreted as a probability density describing the angular redistribution associated with a single elementary scattering event.

Multiple scattering is described as the repeated application of an elementary isotropic scattering via convolution. For two isotropic profiles $f(Q)$ and $g(Q)$, their convolution in transverse momentum space is defined as

$$(f * g)(\mathbf{Q}) \equiv \int_0^\infty q dq \int_0^{2\pi} d\Phi f(q) \times g(\sqrt{Q^2 + q^2 - 2Qq \cos \Phi}). \quad (4)$$

where Φ denotes the relative azimuthal angle between $\mathbf{Q}$ and $\mathbf{q}$. The validity of Eq. (4) relies on the small-angle approximation, under which scattering occurs predominantly in the forward direction.⁴

A key property of this construction is that normalization is preserved under convolution. Specifically, if

$$\int_0^\infty 2\pi Q f(Q) dQ = 1 \quad \text{and} \quad \int_0^\infty 2\pi Q g(Q) dQ = 1, \quad (5)$$

then

$$\int_0^\infty 2\pi Q (f * g)(Q) dQ = 1. \quad (6)$$

By induction, all higher-order self-convolutions preserve this normalization.

The measured (apparent) intensity is therefore obtained as a superposition of contributions from all possible numbers of scattering events,

$$I^{\text{App}}(Q) = P_0 I_0 \delta(Q) + \sum_{n=1}^\infty P_n \frac{\hat{I}^{(*n)}(Q)}{I_0^{n-1}}, \quad (7)$$

where $\hat{I}^{(*n)}$ denotes the n -fold convolution of the single-scattering intensity $\hat{I}(Q)$, with $\hat{I}^{(*1)}(Q) = \hat{I}(Q)$. Here $\delta(Q)$ is normalized with respect to the two-dimensional isotropic measure, such that $\int_0^\infty 2\pi Q \delta(Q) dQ = 1$. The first term corresponds to the unscattered beam and is excluded from the subsequent analysis in the following sections.

A. Algebraic Representation of Isotropic Convolution

In order to express the convolution integral in a mathematically tractable form, we expand the isotropic intensity profile $\hat{I}(Q)$ in an orthonormal basis adapted to the natural Hilbert-space structure induced by the transverse momentum measure. Specifically, we regard the space of square-integrable isotropic functions with weight $2\pi Q dQ$ as a weighted L^2 space and introduce a complete orthonormal set $\{u_n(Q)\}$ in the sense of standard Hilbert-space theory¹³. The basis functions are required to satisfy the orthonormality condition

$$\langle u_n, u_m \rangle \equiv \int_0^\infty 2\pi Q u_n(Q) u_m(Q) dQ = \delta_{nm}, \quad (8)$$

where δ_{nm} denotes the Kronecker delta.

We expand the single-scattering intensity as $\hat{I}(Q) = I_0[\sum_k \alpha_k u_k(Q)]$. This allows the convolution operation to be represented algebraically. The convolution of two basis functions is expressed as a linear combination of the basis set itself,

$$(u_n * u_m)(Q) = \sum_\ell C_{nm}^\ell u_\ell(Q). \quad (9)$$

To evaluate the convolution coefficients C_{nm}^ℓ efficiently, it is advantageous to exploit the isotropy of the problem and work in a representation in which convolution simplifies. For isotropic functions defined on the transverse momentum space, the natural transform is the zeroth-order Hankel transform,

$$\tilde{f}(k) \equiv \mathcal{H}_0[f](k) = \int_0^\infty Q f(Q) J_0(kQ) dQ, \quad (10)$$

with the inverse transform given by

$$f(Q) = \int_0^\infty k \tilde{f}(k) J_0(kQ) dk, \quad (11)$$

where J_0 denotes the Bessel function of the first kind of order zero.

Under this transform, isotropic convolution in two dimensions reduces to multiplication in the transformed domain. Specifically, for two isotropic functions $f(Q)$ and $g(Q)$,

$$(f * g)(Q) = 2\pi \int_0^\infty k \tilde{f}(k) \tilde{g}(k) J_0(kQ) dk. \quad (12)$$

In addition, the Hankel transform satisfies a Parseval-type identity¹⁴,

$$\int_0^\infty Q f(Q) g(Q) dQ = \int_0^\infty k \tilde{f}(k) \tilde{g}(k) dk, \quad (13)$$

which implies, under our definition of the inner product,

$$\langle f, g \rangle = \int_0^\infty 2\pi Q f(Q) g(Q) dQ = 2\pi \int_0^\infty k \tilde{f}(k) \tilde{g}(k) dk. \quad (14)$$

By projecting the convolution expansion onto the basis function u_ℓ , we determine the coefficients C_{nm}^ℓ :

$$\begin{aligned} C_{nm}^\ell &= \langle u_\ell, u_n * u_m \rangle \\ &= \int_0^\infty 2\pi Q u_\ell(Q) \left[2\pi \int_0^\infty k \tilde{u}_n(k) \tilde{u}_m(k) J_0(kQ) dk \right] dQ. \end{aligned} \quad (15)$$

Using the Parseval identity¹³ to resolve the outer integral, this simplifies to a single integral in the transform domain:

$$C_{nm}^\ell = (2\pi)^2 \int_0^\infty k \tilde{u}_\ell(k) \tilde{u}_n(k) \tilde{u}_m(k) dk, \quad (16)$$

where $\tilde{u}_n(k) = \mathcal{H}_0[u_n](k)$ denotes the Hankel transform of the basis function $u_n(Q)$.

With the structure constants C_{nm}^ℓ established, the convolution of any two arbitrary isotropic functions expanded in the basis can be computed algebraically via discrete tensor contraction. Consider two functions $f(Q)$ and $g(Q)$ with expansion coefficients $\{f_n\}$ and $\{g_m\}$, respectively:

$$f(Q) = \sum_n f_n u_n(Q), \quad g(Q) = \sum_m g_m u_m(Q). \quad (17)$$

For comparison, their linear superposition is given simply by

$$(f + g)(Q) = \sum_\ell (f_\ell + g_\ell) u_\ell(Q). \quad (18)$$

In contrast, substituting the expansions of f and g into the bilinear convolution operator yields the convolved function

$$(f * g)(Q) = \sum_\ell h_\ell u_\ell(Q), \quad (19)$$

with expansion coefficients

$$h_\ell = \sum_{n,m} C_{nm}^\ell f_n g_m. \quad (20)$$

This algebraic relation replaces the computationally expensive double integration over the scattering plane with a weighted summation over fixed orthonormal bases.

This formalism extends naturally to the convolution of three or more profiles, such as $(f * g * h)(Q)$. Due to the associativity of convolution, this triple product may be evaluated sequentially. One first computes the intermediate coefficients for $v = f * g$, and subsequently convolves the result with h to obtain $w = v * h$. In terms of the basis coefficients, this two-step process reads

$$v_k = \sum_{i,j} C_{ij}^k f_i g_j, \quad w_\ell = \sum_{k,p} C_{kp}^\ell v_k h_p. \quad (21)$$

The generalization to an arbitrary number of functions follows the same iterative scheme.

In the specific context of the multiple scattering series derived in Eq. (7), we require the n -fold self-convolution of the single-scattering intensity $\hat{I}(Q)$. Let $\{\alpha_k^{(1)}\}$ denote the expansion coefficients of the normalized primary scattering profile $J(Q) \equiv \hat{I}(Q)/I_0$ in the basis $\{u_k\}$,

$$J(Q) = \sum_k \alpha_k^{(1)} u_k(Q). \quad (22)$$

The coefficients $\{\alpha_\ell^{(n)}\}$ corresponding to the n -fold convolution $J^{(*n)}(Q)$ can be determined via the recursive relation

$$\alpha_\ell^{(n)} = \sum_{j,k} C_{jk}^\ell \alpha_j^{(1)} \alpha_k^{(n-1)}, \quad \text{for } n \geq 2, \quad (23)$$

with the base case defined by the initial expansion coefficients of the model function.

To illustrate the practical application of this recursive scheme, we explicitly derive the expansion coefficients for the lowest-order multiple scattering terms.

For the case of double scattering ($n = 2$), the intensity distribution corresponds to the self-convolution of the normalized primary scattering profile, $J(Q)^{(*2)} = J * J$. Its expansion coefficients, denoted by $\{\alpha_\ell^{(2)}\}$, are obtained directly from the self-contraction of the primary coefficients $\{\alpha_k^{(1)}\}$ mediated by the structure constants:

$$\alpha_\ell^{(2)} = \sum_{j,k} C_{jk}^\ell \alpha_j^{(1)} \alpha_k^{(1)}. \quad (24)$$

Here, the structure constant C_{jk}^ℓ dictates how the overlap of basis functions u_j and u_k projects onto the basis function u_ℓ . Computationally, this operation represents a contraction of the pre-calculated rank-3 tensor $\mathbf{C}$ with the vector $\mathbf{a}^{(1)}$ twice.

Proceeding to triple scattering ($n = 3$), the intensity distribution arises from the convolution of the primary profile with the double scattering profile, $J^{(*3)} = J * J^{(*2)}$. Utilizing the recursive definition, the coefficients $\{\alpha_\ell^{(3)}\}$ are calculated by convolving the primary coefficients $\{\alpha_j^{(1)}\}$ with the double scattering coefficients $\{\alpha_k^{(2)}\}$ derived in the previous step:

$$\alpha_\ell^{(3)} = \sum_{j,k} C_{jk}^\ell \alpha_j^{(1)} \alpha_k^{(2)}. \quad (25)$$

If one wishes to express this solely in terms of the primary coefficients, the expression for $\alpha_k^{(2)}$ may be substituted into the sum, revealing the nested structure of the convolution series:

$$\alpha_\ell^{(3)} = \sum_{j,k} C_{jk}^\ell \alpha_j^{(1)} \left(\sum_{p,q} C_{pq}^k \alpha_p^{(1)} \alpha_q^{(1)} \right). \quad (26)$$

This nested form highlights the efficiency of the recursive approach: rather than performing a complex triple contraction over four indices ($j, p, q \rightarrow \ell$) simultaneously, one computes and stores the intermediate vector $\mathbf{a}^{(2)}$, reducing the problem to a sequence of standard matrix-vector-like operations.

Consequently, the total apparent intensity $I^{\text{App}}(Q)$ can be reconstructed by summing the basis functions weighted by these iteratively computed coefficients. Substituting the expansion back into the Poisson summation yields

$$I^{\text{App}}(Q) = I_0 \left[\sum_\ell \beta_\ell u_\ell(Q) + P_0 \delta(Q) \right]. \quad (27)$$

$$\beta_\ell = \sum_{n=1}^{\infty} P_n \alpha_\ell^{(n)}. \quad (28)$$

This result demonstrates the computational efficiency of the basis set approach. Once the structure constants C_{nm}^ℓ are pre-calculated for the universal basis, the evaluation of multiple

scattering effects for any specific model $\hat{I}(Q)$ reduces to simple matrix-vector multiplications, avoiding the need for repeated numerical Hankel transforms.

Given a measured apparent spectrum $I^{\text{App}}(Q)$, the corresponding single-scattering coefficients can be determined by inverting the spectral representation of Eq. (27). After excluding the unscattered $\delta(Q)$ contribution and normalizing by I_0 , the apparent profile is projected onto the orthonormal basis to obtain the coefficients $\{\beta_\ell^{\text{Obs}}\}$. Using the relation in Eq. (28), these observed coefficients are expressed as nonlinear functions of the unknown primary coefficients $\{\alpha_\ell^{(1)}\}$ through the recursively generated $\{\alpha_\ell^{(n)}\}$.

The inversion may be formulated as a maximum-likelihood estimation (MLE) problem in intensity space. Let $I^{\text{Obs}}(Q_i)$ denote the experimentally measured apparent intensity at discrete momentum-transfer points Q_i , and let $I^{\text{App}}(Q_i; \{\alpha_\ell^{(1)}\})$ denote the model prediction, parameterized through the Poisson-weighted multiple-scattering series. Assuming independent measurement with known variances ΔI_i^2 , the log-likelihood function is

$$\log \mathcal{L}(\{\alpha_\ell^{(1)}\}) = -\frac{1}{2} \sum_i \frac{[I^{\text{Obs}}(Q_i) - I^{\text{App}}(Q_i; \{\alpha_\ell^{(1)}\})]^2}{\Delta I_i^2} + \text{const}. \quad (29)$$

Maximization of $\mathcal{L}$ is therefore equivalent to a weighted least-squares minimization of the residual between the measured and predicted intensities. In practice, because the Poisson weights P_n decay rapidly with increasing n , higher-order scattering events carry progressively negligible probability mass. The multiple-scattering series may therefore be truncated at a finite order with minimal impact on the predicted intensity. From a statistical perspective, this truncation may be interpreted either as neglecting events with vanishing prior probability or, equivalently, as conditioning on a restricted range of scattering orders. In both interpretations, the rapid decay of P_n ensures that the dominant contribution arises from low-order events, enabling stable and well-posed recovery of the primary single-scattering component.

B. Construction of an Orthonormal Expansion Basis

In this section, we construct an explicit orthonormal basis for isotropic scattering profiles that is compatible with the two-dimensional Hankel geometry established in the previous section. The reduced single-event spectrum $J(Q) = \hat{I}(Q)/I_0$ is nonnegative, regular at $Q = 0$, and decays at large Q . We therefore begin from a smooth, one-parameter family of admissible functions that capture these properties.

We introduce the Lorentzian-type family

$$\psi_b(Q) \equiv \frac{1}{(1 + a^2 Q^2)^b}, \quad b > 1, \quad (30)$$

where a is a characteristic length scale. The condition $b > 1$ ensures integrability of Eq. (5). These functions are analytic at

$Q = 0$ and decay algebraically at $Qa \gg 1$, making them suitable as building blocks for isotropic scattering intensities. In practice, we restrict b to half-integer values in order to obtain closed-form Hankel transforms and simplified expressions for the associated modified Bessel functions. Consequently, the smallest admissible practical choice is $b = 3/2$.

The first step is to derive the Hankel transform of $\psi_b(Q)$ explicitly. Using the property of Gamma function

$$(1+z)^{-b} = \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} e^{-t} e^{-tz} dt, \quad \Re(b) > 0, \quad (31)$$

with $z = a^2 Q^2$, we write

$$\psi_b(Q) = \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} e^{-t} e^{-a^2 t Q^2} dt. \quad (32)$$

Substituting into the Hankel transform and exchanging the order of integration yields

$$\begin{aligned} \tilde{\psi}_b(k) &= \int_0^\infty Q \psi_b(Q) J_0(kQ) dQ \\ &= \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} e^{-t} \left[\int_0^\infty Q e^{-a^2 t Q^2} J_0(kQ) dQ \right] dt. \end{aligned} \quad (33)$$

The inner integral follows the standard Bessel–Gaussian identity^{15,16}

$$\int_0^\infty Q e^{-\beta Q^2} J_0(kQ) dQ = \frac{1}{2\beta} \exp\left(-\frac{k^2}{4\beta}\right), \quad \Re(\beta) > 0, \quad (34)$$

with $\beta = a^2 t$. Therefore,

$$\tilde{\psi}_b(k) = \frac{1}{2a^2 \Gamma(b)} \int_0^\infty t^{b-2} \exp\left(-t - \frac{k^2}{4a^2 t}\right) dt. \quad (35)$$

The next task is to handle with the t -integral. The integral identity of the modified Bessel function K_ν shows¹⁷,

$$\int_0^\infty t^{\nu-1} \exp\left(-t - \frac{z^2}{4t}\right) dt = 2 \left(\frac{z}{2}\right)^\nu K_\nu(z). \quad (36)$$

K_ν is defined as

$$K_\nu(z) = \frac{\pi}{2} \frac{I_{-\nu}(z) - I_\nu(z)}{\sin(\nu\pi)} \quad (37)$$

$$I_\nu(z) = \sum_{m=0}^\infty \frac{1}{m! \Gamma(m+\nu+1)} \left(\frac{z}{2}\right)^{2m+\nu} \quad (38)$$

With $\nu = b-1$ and $z = k/a$, we obtain

$$\tilde{\psi}_b(k) = \frac{1}{2^{b-1} a^2 \Gamma(b)} \left(\frac{k}{a}\right)^{b-1} K_{b-1}\left(\frac{k}{a}\right). \quad (39)$$

It is worth noting a correspondence between Eq. (39) and the Matérn covariance family commonly used in spatial statistics^{18,19}. The Matérn covariance function is defined as

$$\rho_\nu(r) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{r}{\ell}\right)^\nu K_\nu\left(\frac{r}{\ell}\right), \quad (40)$$

where $\nu > 0$ and ℓ is a correlation length. Comparing Eqs. (39) and (40), the zeroth-order Hankel transform of the Lorentzian-type spectrum $\psi_b(Q) = (1 + a^2 Q^2)^{-b}$ is seen to coincide with the Matérn form in the transform variable k , up to normalization¹⁹, with the identification $\nu = b-1$ and $\ell = a$.

Define the variable x in compact space

$$x \equiv \frac{1}{1 + a^2 Q^2}, \quad x \in (0, 1], \quad (41)$$

so that the Lorentzian seed function corresponds to $\psi_b(Q) = x(Q)^b$. The Jacobian of this transformation follows directly as

$$dx = -2a^2 Q (1 + a^2 Q^2)^{-2} dQ = -2a^2 Q x^2 dQ, \quad (42)$$

which implies

$$Q dQ = -\frac{1}{2a^2} x^{-2} dx. \quad (43)$$

We restrict our attention to basis functions on the scattering plane that differ from the seed function by a polynomial modulation in the compact variable x . Substituting trial functions of the form $f(Q) = x(Q)^b f_x(x)$ and $g(Q) = x(Q)^b g_x(x)$ into the isotropic inner product (Eq. (14)) and applying the change of variables yields, up to a constant prefactor,

$$\langle f, g \rangle \propto \int_0^1 x^{2b-2} f_x(x) g_x(x) dx. \quad (44)$$

This structure motivates the construction of orthonormal basis functions

$$u_n(Q) = x(Q)^b p_{n-1}(x(Q)), \quad n = 1, 2, \dots, \quad (45)$$

where $\{p_n\}$ are polynomials orthonormal on $[0, 1]$ with respect to the weight $w(x) = x^{2b-2}$. We identify these as the shifted Jacobi family¹⁷

$$p_n(x) = \sqrt{2n+2b-1} P_n^{(2b-2, 0)}(1-2x), \quad (46)$$

which satisfies

$$\int_0^1 x^{2b-2} p_n(x) p_m(x) dx = \delta_{nm}. \quad (47)$$

Here $P_n^{(\alpha, \beta)}(t)$ denotes the standard Jacobi polynomial of degree n with parameters $\alpha = 2b-2$ and $\beta = 0$ ²⁰

$$P_n^{(\alpha, \beta)}(t) = \sum_{k=0}^n \binom{n+\alpha}{n-k} \binom{n+\beta}{k} \left(\frac{t-1}{2}\right)^k \left(\frac{t+1}{2}\right)^{n-k}. \quad (48)$$

Explicitly, we define the orthonormal basis functions on the scattering plane by shifting the index so the sequence begins at $n = 1$. The resulting expression is

$$u_n(Q) = \sqrt{\frac{a^2}{\pi}} x(Q)^b \sqrt{2n+2b-3} P_{n-1}^{(2b-2, 0)}(1-2x(Q)), \quad (49)$$

for $n = 1, 2, \dots$. The prefactor $\sqrt{a^2/\pi}$ is included to ensure normalization with respect to the physical area element $2\pi Q dQ$.

To evaluate the Hankel transform, we expand the polynomial part $p_{n-1}(x)$ of degree $n-1$ as

$$p_{n-1}(x) = \sum_{j=0}^{n-1} d_{n,j} x^j. \quad (50)$$

For convenience, the coefficients $d_{n,j}$ are written explicitly as

$$d_{n,j} = \sqrt{2n+2b-3} \sum_{k=0}^j (-1)^k \times \binom{n+2b-3}{n-1-k} \binom{n-1}{k} \binom{n-1-k}{j-k}. \quad (51)$$

Substituting this expansion into Eq. (45) allows us to express the basis function as a finite linear combination of Generalized Lorentzians:

$$u_n(Q) = \sum_{j=0}^{n-1} d_{n,j} x(Q)^{b+j} = \sum_{j=0}^{n-1} c_{n,j} \frac{1}{(1+a^2 Q^2)^{b+j}}, \quad (52)$$

where the combined coefficients are

$$c_{n,j} = \sqrt{\frac{a^2}{\pi}} \sqrt{2n+2b-3} d_{n,j}. \quad (53)$$

Since the Hankel transform of each Lorentzian term $\psi_{b+j}(Q)$ is known analytically from Eq. (39), the Hankel transform of the full basis function $u_n(Q)$ follows immediately by linearity:

$$\tilde{u}_n(k) = \sum_{j=0}^{n-1} \frac{c_{n,j}}{2^{b+j-1} a^2 \Gamma(b+j)} \left(\frac{k}{a}\right)^{b+j-1} K_{b+j-1}\left(\frac{k}{a}\right). \quad (54)$$

Fig. 1 illustrates representative members of the orthonormal basis $\{u_n(Q)\}$ for $b = 3/2$ constructed from a Lorentzian-type seed family, together with their corresponding zeroth-order Hankel transforms $\{\tilde{u}_n(k)\}$ given analytically by Eq. (54), following a strategy similar to recent orthonormal basis expansions developed for small-angle scattering analysis²¹, which construct orthonormal bases compatible with the physical boundary conditions of isotropic scattering while avoiding reliance on explicit structural models.

Panel (a) and (c) shows the reciprocal-space basis functions $u_n(Q)$ for the lowest orders n , plotted as functions of the dimensionless variable $A = aQ$, highlighting their regular behavior at $Q = 0$, compatibility with the physical boundary conditions of isotropic scattering profiles, and controlled algebraic decay at large Q . The increasing number of sign changes with n reflects the polynomial modulation imposed by the shifted Jacobi polynomials, while exact orthonormality is preserved with respect to the isotropic measure $2\pi Q dQ$, ensuring a well-conditioned spectral representation.

Panel (b) and (d) displays the corresponding Hankel-transformed basis functions $\tilde{u}_n(k)$ as functions of the scaled transform variable $B = k/a$, demonstrating that each basis element in transformed variable admits a closed-form spectral representation in terms of modified Bessel functions of the second kind. The rapid decay of $\tilde{u}_n(k)$ at large k , together with their smooth behavior near the origin governed by the

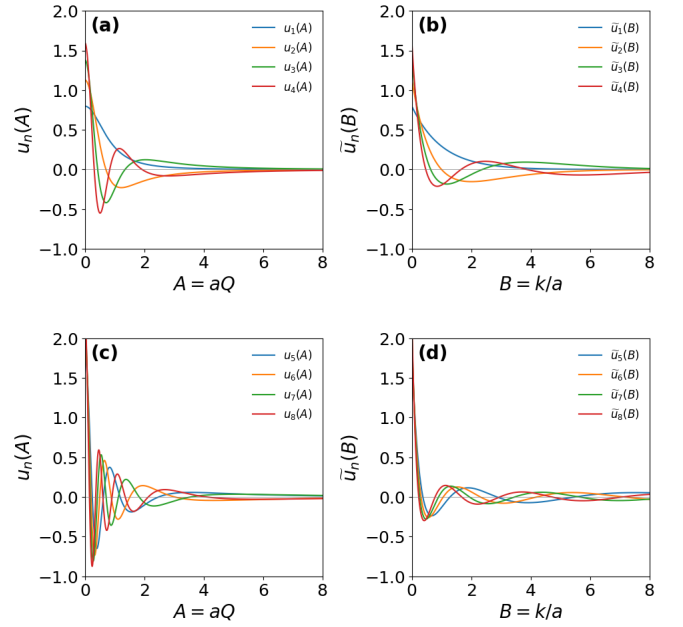


FIG. 1. Orthonormal basis functions constructed from the Lorentzian-type family and their corresponding zeroth-order Hankel transforms. (a), (c) Real-space basis functions $u_n(Q)$ plotted as functions of the scaled variable $A = aQ$. (b), (d) Hankel-transformed basis functions $\tilde{u}_n(k)$ shown as functions of $B = k/a$.

parameter b , ensures numerical stability and efficiency when evaluating convolution coefficients and higher-order multiple-scattering contributions in the transform domain by Eq. (16).

The reciprocal- and transformed-space representations in Fig. 1 make clear that the constructed basis is orthonormal, analytically tractable, and intrinsically suited to isotropic convolution operators in two-dimensional scattering geometries.

III. NUMERICAL BENCHMARK

To assess the reliability of the truncated spectral representation under repeated convolution, we examine how the truncated basis expansion in Eq. (22) approximates a normalized profile $J(Q)$ and its successive self-convolutions. The primary intensity is written as $\hat{I}(Q) = I_0 J(Q)$ with normalization defined in Eq. (3). In practice, the expansion is truncated to the first L basis functions, and both the primary spectrum and its convolved forms are represented within the same L -dimensional subspace. As a representative example, we consider the fuzzy-ball scattering function

$$\hat{I}(Q) = 9 \frac{[\sin(QR) - QR \cos(QR)]^2}{(QR)^6} \exp\left(-\frac{(Qs)^2}{2}\right), \quad (55)$$

with $R = 2$ and $s = 0.5$. This expression corresponds to the product of the homogeneous hard-sphere form factor^{2,22},

$$\left[\frac{3(\sin(QR) - QR \cos(QR))}{(QR)^3} \right]^2,$$

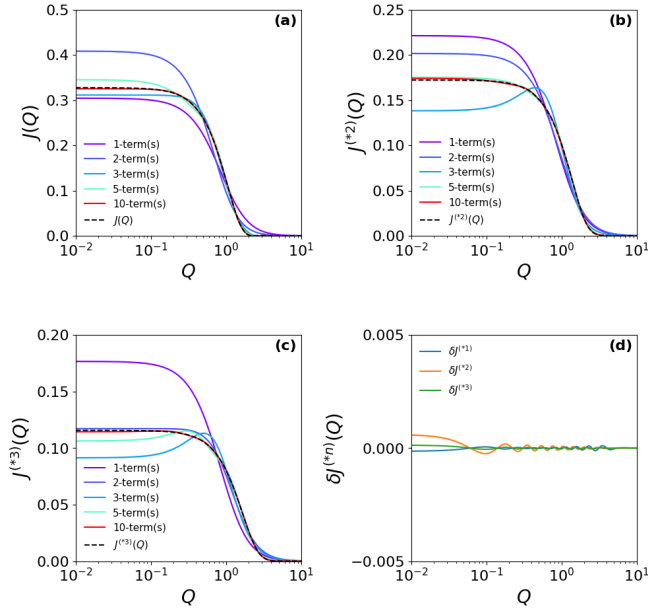


FIG. 2. Convergence of the truncated spectral representation under repeated isotropic convolution. Panels (a)–(c) show the reconstructed spectra $J(Q)$, $J^{(2)}(Q)$, and $J^{(3)}(Q)$ obtained with increasing truncation order L . The reference curve corresponds to a sufficiently large basis size. Panel (d) displays the deviation $\delta J^{(n)}(Q) = J_L^{(n)}(Q) - J_{\text{Ref}}^{(n)}(Q)$.

and a Gaussian damping factor $\exp[-(Qs)^2/2]$, which phenomenologically accounts for interfacial roughness or a diffuse corona surrounding the core. The parameter R represents the core radius, while s characterizes the degree of surface smearing, thereby attenuating the high- Q oscillations associated with a sharp hard-sphere boundary.

Panels (a)–(c) of Fig. 2 show the reconstructed single-, double-, and triple-scattering contributions, $J(Q)$, $J^{(2)}(Q)$, and $J^{(3)}(Q)$, for increasing truncation levels L . The convolved spectra approach the reference solution rapidly as L increases. Panel (d) quantifies the deviation

$$\delta J^{(n)}(Q) = \sum_{\ell=1}^L \alpha_{\ell}^{(n)} u_{\ell}(Q) - J_{\text{Ref}}^{(n)}(Q), \quad (56)$$

where $J_{\text{Ref}}^{(n)}(Q)$ is obtained by numerical integration. For the first three convolution orders, which provide the dominant contributions in Eq. (7), the deviations remain small, indicating that the truncated basis captures the convolved profiles with sufficient accuracy for evaluating the Poisson-weighted multiple-scattering intensity.

To examine the behavior of the convolution algebra in a finite-dimensional orthonormal space, we analyze the coefficients generated under truncation of the rank-3 structure tensor. In practice, the tensor defined in Eqs. (15) and (16) is restricted to its leading $L \times L \times L$ block, and the convolution coefficients are computed via Eq. (24) using only indices $1 \leq j, k, \ell \leq L$. As a representative case, we focus on the double-scattering coefficients, which provide the first nontriv-

ial test of closure within the truncated basis. This restriction confines the convolution operation to the truncated subspace and enables direct manipulation at the level of expansion coefficients, without requiring numerical integration.

Fig. 3(a) presents a heat map of the deviation in the expansion coefficients for $J^{(2)}$,

$$\delta \alpha_{\ell}^{(2)}(L) = \alpha_{\ell}^{(2)}(L) - \alpha_{\ell, \text{Ref}}^{(2)}, \quad (57)$$

as a function of truncation order L and coefficient index ℓ , where $\alpha_{\ell, \text{Ref}}^{(2)}$ is obtained directly via numerical integration. The color scale distinguishes positive and negative deviations, while gray regions correspond to coefficients not defined at a given truncation level. As L increases, the effect of truncation remains confined to the higher-order components, whereas the leading coefficients are stably and accurately determined.

Fig. 3(b) translates this coefficient-level deviation into physical space by displaying the reconstruction error in the double-scattering spectrum,

$$\delta J^{(2)}(Q) = J_L^{(2)}(Q) - J_{\text{Ref}}^{(2)}(Q), \quad (58)$$

where $J_L^{(2)}(Q)$ is computed at truncation level L and $J_{\text{Ref}}^{(2)}(Q)$ is obtained via numerical integration. As L increases, the magnitude of $\delta J^{(2)}(Q)$ decreases systematically, consistent with the stabilization of the leading expansion coefficients and the confinement of truncation effects to higher-order modes.

These results indicate that the implementation of the tensor contraction defined by Eqs. (16) and (24) remains numerically well behaved when carried out in a truncated spectral space. Restricting the representation to a finite-dimensional subspace does not introduce artificial long-wavelength distortions. It is thus sufficient to evaluate the dominant contributions to the Poisson-weighted multiple-scattering series in terms of the single-scattering expansion coefficients within this truncated basis.

Having examined the stability of the convolution algebra, we next demonstrate the feasibility of the proposed desmearing framework in a forward-inverse setting. Fig. 4 shows a test in which a known primary scattering intensity $\hat{I}(Q)$ is subjected to multiple scattering according to Eq. (7) with transmission $T = 0.8, 0.25$ and 0.1 , generating an apparent intensity $I_{\text{App}}(Q)$ used as synthetic observed data.

The black dashed curve denotes the true primary intensity $\hat{I}(Q)$, while the gray symbols represent the apparent spectrum $I_{\text{App}}(Q)$ distorted by multiple scattering. For visual comparison, the primary intensity is scaled by $(1 - T)$, so that $I_{\text{App}}(Q)$ coincides with the scaled $\hat{I}(Q)$ in the limit $T \rightarrow 1$. The repeated self-convolution results in noticeable broadening of the intensity profile relative to the primary signal.

The red curve shows the least square fit of $I_{\text{App}}(Q)$ obtained by expanding the unknown reduced spectrum $J(Q) = \hat{I}(Q)/I_0$ in the orthonormal basis $\{u_{\ell}(Q)\}$ and propagating the coefficients through the recursive convolution relations in Eqs. (23)–(26), followed by Poisson weighting via Eq. (28). The excellent agreement between the red curve and the gray symbols confirms that the truncated spectral representation accurately captures the nonlinear convolution structure induced by multiple scattering.

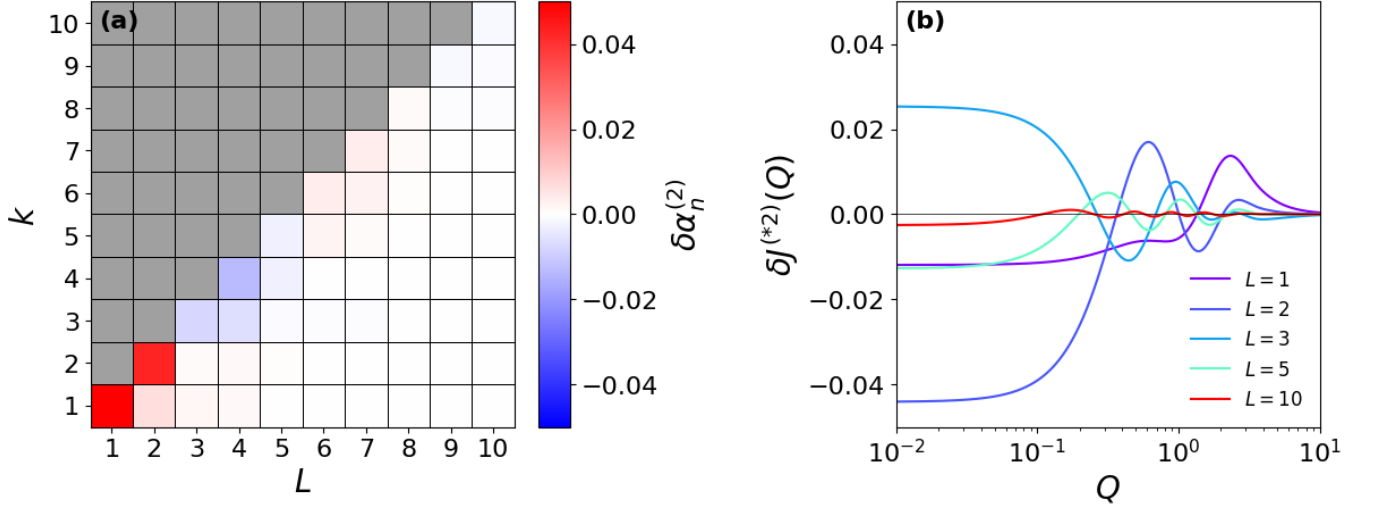


FIG. 3. The expansion coefficients of the convolved spectrum are evaluated using a finite C_{nm}^ℓ tensor of size $L \times L \times L$. (a) Heat map of the coefficient deviation as a function of truncation order L and coefficient index k . Positive and negative deviations are rendered in red and blue, respectively, while gray regions indicate coefficients that are not defined for a given truncation. (b) Corresponding deviation $\delta I^{(*)2}(Q)$ of the reconstructed double-scattering intensity for selected truncation levels L , showing convergence of the finite-dimensional representation.

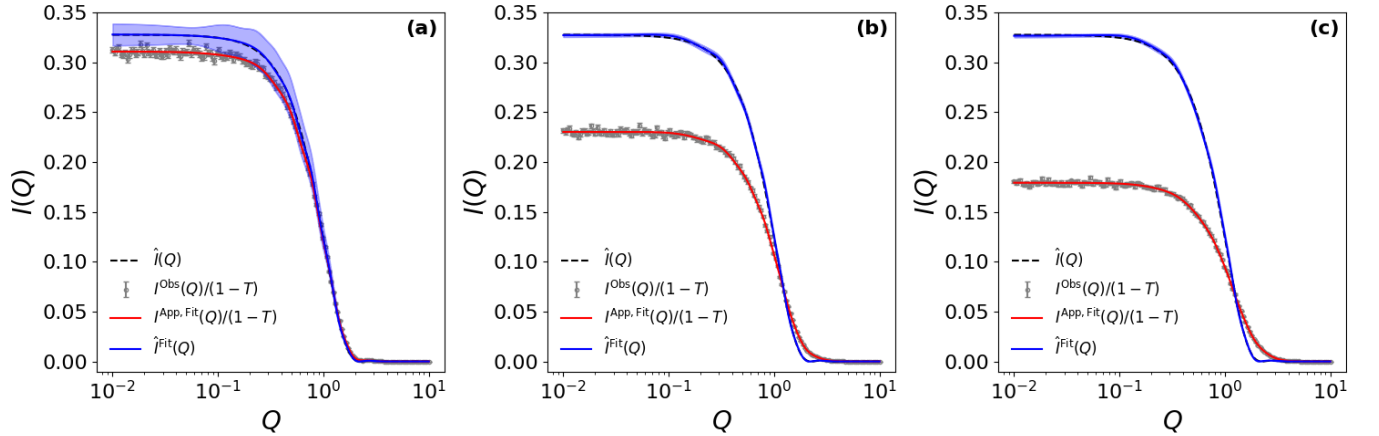


FIG. 4. Demonstration of multiple-scattering correction using the proposed spectral framework. The black dashed curve denotes the true primary scattering intensity, whereas the gray symbols indicate the apparent intensity distorted by different degrees of multiple scattering (panel (a) $T = 0.8$, (b) $T = 0.25$ and (c) $T = 0.1$). The red curve denotes the forward model fit to the distorted data, and the blue curve shows the reconstructed primary spectrum obtained from the inferred spectral coefficients. The excellent agreement between the reconstructed and true primary intensities demonstrates the accuracy and stability of the proposed desmearing algorithm.

Most importantly, the reconstructed primary spectrum $\hat{I}^{\text{Fit}}(Q)$, shown as the blue curve, is obtained directly from the optimized coefficients $\alpha_\ell^{(1)}$. Its near-perfect overlap with the true primary curve (black dashed) demonstrates that the inverse mapping from $I_{\text{App}}(Q)$ to $\hat{I}(Q)$ is stable and quantitatively accurate within the constructed orthonormal basis. Although the reconstructed primary spectra in Figs. 4(a)–4(c) do not coincide exactly, their deviations from the ground truth, normalized by $I(0)$, remain at the level of $\sim 1\%$, comparable to the measurement noise.

This result verifies that the convolution operator defined in Eq. (4), when represented in the finite dimensional expansion form of Eqs. (16) and (24), still admits a numerically

well-conditioned inverse under Poisson weighted combination. In particular, the truncation does not introduce notable artificial oscillations, thereby enabling robust recovery of the single-scattering intensity without explicit integral transforms or model-dependent extrapolations.

IV. APPLICATION TO EXPERIMENTAL SANS DATA

While the preceding numerical benchmarks establish the stability and invertibility of the proposed framework under controlled conditions, a decisive test requires application to real experimental data in which multiple scattering is intrinsic.

sically present. We therefore apply the algebraic desmearing procedure to SANS measurements collected at various neutron scattering facilities for several representative soft-matter and nanostructured systems exhibiting substantial multiple scattering.

Fig. 5 presents experimental SANS data for (a) quantum dots, (b) nanoclays, (c) aggregates of polyethylene glycol-block-alkyne functional polylactide-block-polylactide (PEG-b-PLA_{kyne}-b-PLA), (d) lysozyme ecocorona nanoplastics–soil system, (e) a film of dried gypsum, and (f) a film of wet gypsum. Panels (a)–(c) were measured at the EQ-SANS instrument at the Spallation Neutron Source (SNS), panel (d) at the CG3 SANS instrument at the High Flux Isotope Reactor (HFIR), and panels (e)–(f) at the D22 SANS instrument at the Institut Laue-Langevin (ILL). In all cases, the gray symbols denote the experimentally measured scattering intensity $I(Q)$ affected by multiple scattering.

The corresponding transmission values T for panels (a)–(f) are 0.67, 0.68, 0.69, 0.74, 0.10, and 0.042, respectively. Here, T collectively accounts for neutron transmission and absorption within the sample, following the definition introduced by Refs. 4 and 7. We emphasize that the transmission T used in this work corresponds to the value routinely reported by instrument scientists and is therefore the quantity most directly accessible to users. Rather than attempting to disentangle absorption and scattering contributions at a microscopic compositional level, we adopt this overall transmission as part of experimental calibration. This choice is particularly appropriate for complex soft-matter and environmental systems, where the detailed distribution of absorbing species is often not known with sufficient precision to justify a fully microscopic decomposition of various attenuation effects.

The red curves correspond to the forward fits obtained by propagating the coefficients of primary spectrum through the Poisson-weighted convolution series described in Sec. II. The quantitative agreement between the red curves and the gray symbols confirms that the recursive formulation captures the nonlinear distortion induced by multiple scattering across diverse structural morphologies and instruments.

The blue curves represent the reconstructed primary scattering intensities obtained from the inferred first-order expansion coefficients. For panels (a)–(d), which span a range of soft-matter and complex environmental systems with moderate transmission values, the reconstructed spectra exhibit physically consistent trends and recover smooth low- Q behavior consistent with the expected redistribution of scattered intensity toward the forward direction (lower scattering angles) induced by correction of multiple scattering. The vertical dotted line marks the characteristic scale $1/a$ associated with the basis parameter, indicating the Q region over which variations in the expansion coefficients most strongly influence the reconstruction. Across this crossover region, the corrected profiles remain smooth and free of artificial oscillations.

Panels (e) and (f) represent particularly stringent tests of the method, with extremely low transmission values of $T = 0.10$ (dried gypsum) and $T = 0.042$ (wet gypsum). Such strongly attenuating samples correspond to severe multiple-scattering conditions under which conventional desmearing approaches,

particularly those relying on numerical Fourier transforms and logarithmic inversion do not guarantee numerical stability, as small calibration or statistical errors are strongly amplified⁴. Despite these extreme transmission levels, the algebraic convolution framework still yields stable forward fits and well-behaved reconstructed primary spectra. The absence of spurious oscillations and the physically consistent low- Q behavior demonstrate that the proposed method remains applicable even when total scattering probability approaches unity.

These experimental results indicate that the proposed algebraic framework achieves stable and quantitative correction of multiple-scattering effects across different neutron facilities, wavelength settings, and transmission ranges. By working directly in the spectral domain, the approach avoids explicit integral transforms of truncated data, thereby mitigating Gibbs-type oscillations,⁸ and removing the need for model-dependent extrapolation. This permits reliable structural interpretation even for thick or highly attenuating samples, where conventional correction schemes often become less reliable.

To further illustrate the practical significance of multiple-scattering correction, we examine the problem of merging ultra-small-angle neutron scattering (USANS) and SANS data collected at different wavelengths and transmission conditions. Figure 6 presents this situation for a polymer blend sample.

The left segment of the black dashed curve corresponds to the experimentally measured USANS intensity $I(Q)$ in the range $Q < 10^{-3} \text{ \AA}^{-1}$. For the Bonse-Hart USANS at SNS, it was acquired using the first harmonic with neutron wavelength of 3.6 Å. Because USANS spectra are strongly affected by the slit geometry, the measured intensity in this low- Q regime contains contributions from higher- Q components. Consequently, it must be stitched to slit-smeared SANS data covering $Q > 10^{-3} \text{ \AA}^{-1}$, which serves as the higher- Q reference for slit desmearing.

The desmeared isotropic intensities shown as solid curves in Fig. 6 can only be obtained by combining the stitched black dashed curve over the full USANS–SANS Q -range with the known point-spread function (PSF) of the Bonse–Hart slit geometry through regularized regression¹⁰.

In Fig. 6, synthetic, resolution-free multiply scattered datasets shown as colored solid curves for different transmission values, generated from a reference case with $T = 0.8$, display clear transmission-dependent deviations within the USANS Q -range. Similar distortions are also visible in the corresponding slit-smeared experimental spectra, plotted as colored dashed curves. These transmission-induced discrepancies hinder consistent stitching with the SANS data near $Q \approx 10^{-3} \text{ \AA}^{-1}$ and lead to instability in the removal of the Bonse–Hart slit PSF.

This example highlights that an accurate treatment of transmission within the Poisson-weighted convolution formalism is essential for reliable cross-instrument data matching. In applications where USANS and SANS data must be combined to span several decades in Q , correction of multiple scattering is therefore not merely a refinement but a prerequisite for physically meaningful structural interpretation.

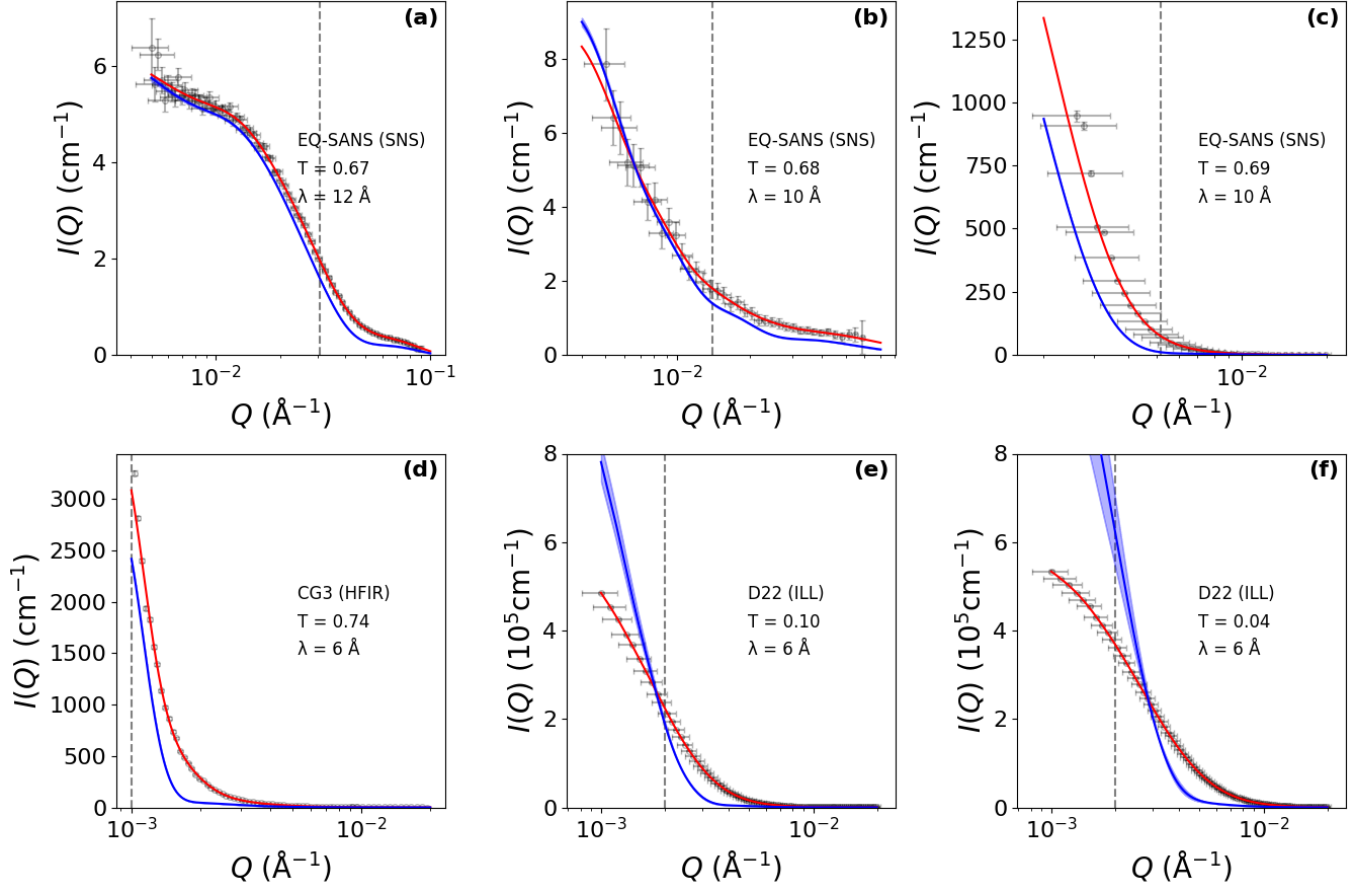


FIG. 5. Experimental demonstration of multiple-scattering correction applied to SANS measurements collected at multiple neutron facilities. Panels (a)–(c) were measured at EQ-SANS (SNS), panel (d) at CG3 SANS (HFIR), and panels (e)–(f) at D22 (ILL). The samples are (a) quantum dots, (b) nanoclays, (c) aggregates of polyethylene glycol-block-alkyne functional polylactide-block-polylactide (PEG-b-PLA_{ne}-b-PLA), (d) lysozyme ecocorona nanoplastics–soil system, (e) film of dried gypsum, and (f) film of wet gypsum. Gray symbols represent the experimentally measured intensity $I(Q)$ affected by multiple scattering. Red curves denote the forward fits obtained from the algebraic convolution model. Blue curves show the reconstructed primary scattering intensity recovered from the inferred expansion coefficients. The vertical dotted line indicates the characteristic scale $1/a$ associated with the basis parameter.

V. RELATION TO EXISTING APPROACHES

The successful reconstruction of the primary spectra in Figs. 4–5 demonstrates that the present algebraic formulation provides a stable and quantitative correction of multiple scattering. It is therefore instructive to compare this framework with existing correction approaches. In some methods, multiple scattering is incorporated through forward calculation based on a prescribed analytical model for the single-scattering intensity, which is effective when such a model is reliably available.¹² By contrast, although the present framework is also built on a chosen basis representation, the expansion coefficients remain free parameters to be determined from the data, while the associated structure coefficients can be precomputed within the selected basis.

Other approaches instead rely on integral transforms of experimental data, as in the MuScatt implementation.⁷ Conceptually, the Poisson-weighted superposition in Eq. (7) corresponds to the iterative differential representation given in

Eq. (2) of Ref. 7, as both describe the measured intensity as a superposition of successive scattering orders generated by repeated application of a single-scattering operator under statistical independence. In the transformed domain, this structure becomes explicit: under the zeroth-order Hankel transform defined in Eqs. (10)–(11), isotropic convolution reduces to multiplication, and summation of the Poisson series yields

$$\widetilde{I^{\text{APP}}}(k) = \frac{P_0 I_0}{2\pi} + \frac{I_0}{2\pi} \left[\exp \left(2\pi\mu \frac{\tilde{I}(k)}{I_0} \right) - 1 \right]. \quad (59)$$

This expression naturally yields Eq. (9) of Ref. 7, where the Hankel-transformed apparent intensity $\widetilde{I^{\text{APP}}}(k)$, denoted as $\tilde{i}(r)$ in Ref. 7, serves as the primary working quantity.

However, experimental SANS data are necessarily confined to a finite Q window. Direct Fourier transformation therefore introduces a sharp truncation at the measurement boundaries, leading to artificial oscillations associated with the Gibbs phenomenon.⁸ To mitigate these truncation arti-

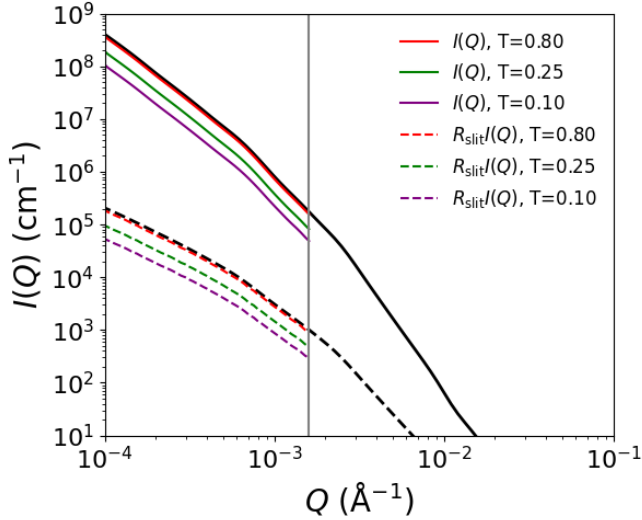


FIG. 6. Importance of multiple-scattering correction for merging USANS and SANS measurements. Dashed curves show the intensity $I(Q)$ of a polymer blend with Bonse-Hart slit smearing. For $Q < 10^{-3} \text{ \AA}^{-1}$, the data can be obtained from USANS. Because slit desmearing requires the full Q -range, the USANS data are stitched to the SANS spectrum that has been artificially slit-smearred for $Q > 10^{-3} \text{ \AA}^{-1}$. Solid curves denote the desmeared intensities by regularized regression. As transmission decreases, the synthetic multiply-scattered profiles (colored solid curves) depart from the $T = 0.8$ reference. Such deviations are already visible in the slit-smearred USANS spectra (colored dashed curves). Proper correction of transmission-dependent multiple scattering is required to achieve consistent overlap with SANS data in the intermediate- Q regime.

facts, real-space implementations typically rely on an analytical model of $I(Q)$ defined over $Q \in (0, \infty)$ to complete the spectrum. While effective when an accurate structural model is available, this strategy inevitably embeds model assumptions into the correction.

In addition, as noted in Ref. 4, inverse procedures based on transform-space logarithmic relations become increasingly sensitive as the total true scattering probability approaches unity. In that regime, reliable inversion requires experimental uncertainties well below the percent level and absolute calibration precision significantly better than the deviation of $T = e^{-\mu}$ from zero; otherwise, small statistical or calibration errors are strongly amplified, and accurate recovery of the single-scattering spectrum cannot be ensured.

In contrast, the present framework avoids both transform-space truncation and logarithmic inversion of globally integrated quantities. The measured spectrum is expanded directly in an orthonormal basis $\{u_\ell(Q)\}$ with analytically known zeroth-order Hankel transforms over $(0, \infty)$, following our previous orthonormal expansion strategy.²¹ Convolution is evaluated algebraically through precomputed structure constants within this spectral representation.

Because the expansion is carried out in an orthonormal basis, the forward multiple-scattering operator and its inversion are represented as finite-dimensional linear mappings acting on coefficient vectors. Recovery of the single-scattering spec-

trum therefore reduces to a well-posed least-squares problem whose numerical behavior is governed by the singular-value structure of the coefficient matrix. No division by a normalization factor approaching zero is involved, and the orthonormal projection preserves norm equivalence. As a result, small perturbations in the measured intensity lead to proportionally small perturbations in the expansion coefficients, provided the operator remains well conditioned. The numerical stability of the procedure is therefore governed by the conditioning of a finite-dimensional coefficient matrix, rather than a scalar singularity tied to the total integrated intensity.

One should be cautious that the present basis construction is not universally optimal. For systems whose scattering intensity exhibits pronounced oscillatory features, such as the sharp minima and cusps characteristic of hard particles²³ or highly faceted geometries including Platonic solids,²⁴ general polyhedral form factors,²⁵ and polygonal or prismatic particles,²⁶ the Lorentzian-type basis developed here may require a substantially larger number of expansion terms to accurately capture these spectral details. In such cases, convergence may become slower, and a different set of orthonormal basis functions, tailored to the inherent boundary conditions and oscillatory structure of the experimentally measured two-point static correlation functions, would be more appropriate.

Thus, while both approaches share the same Born-type physical foundation, the integral-transform-based method attains stability through transformation and model completion,⁷ whereas the present algebraic formulation achieves stability through a complete, model-agnostic spectral representation applied directly to the measured data, with the understanding that alternative bases may be advantageous for strongly oscillatory systems.

VI. CONCLUSIONS

We have developed an algebraic framework for correcting multiple-scattering effects in small-angle neutron scattering (SANS) based on an orthonormal spectral representation of isotropic intensity profiles. By expanding the primary scattering intensity in a basis adapted to the natural transverse-momentum measure, isotropic convolution is reduced to a tensor contraction governed by precomputed structure constants. Within this representation, the Poisson-weighted multiple-scattering series can be evaluated recursively within a finite-dimensional spectral space, eliminating the need for numerical integral transforms and model-dependent extrapolation beyond the measured Q range.

Numerical benchmarks demonstrate systematic convergence under repeated convolution and confirm that truncation errors remain controlled as convolution order increases. Coefficient-level analysis further shows that the convolution operator is well conditioned in the constructed basis, preventing the amplification of high-order modes. A controlled forward-inverse test verifies accurate recovery of the single-scattering intensity from multiply scattered data. Application to SANS measurements exhibiting significant multiple scattering confirms that the distorted experimental spectrum can

be quantitatively reproduced and the underlying primary intensity reconstructed with high fidelity.

The present approach provides a stable, model-agnostic alternative to conventional desmearing methods that depend on integral transforms and analytic tail assumptions. Formulated entirely in a finite spectral representation and implemented through tensor contractions, the method is computationally efficient and naturally extendable to resolution corrections, Bayesian conditioning, and integration with existing SANS data-reduction workflows. More generally, the algebraic framework developed here offers a systematic means of addressing multiple-scattering effects in regimes where traditional correction strategies lose reliability.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A: Explicit low-order basis functions for $b = \frac{3}{2}$

We fix $b = \frac{3}{2}$ and introduce the reduced dimensionless variables

$$A \equiv aQ, \quad B \equiv \frac{k}{a}, \quad x \equiv \frac{1}{1+A^2} \in (0, 1]. \quad (\text{A1})$$

For $b = \frac{3}{2}$, the orthonormal basis functions on the scattering plane are given by

$$u_n(Q) = \frac{a}{\sqrt{\pi}} \sqrt{2n} x^{3/2} P_{n-1}^{(1,0)}(1-2x), \quad (\text{A2})$$

where $P_m^{(\alpha,\beta)}$ are Jacobi polynomials. The first few functions are explicitly:

$$u_1(Q) = \frac{a}{\sqrt{\pi}} \sqrt{2} x^{3/2}. \quad (\text{A3})$$

$$u_2(Q) = \frac{a}{\sqrt{\pi}} 2 \left(2x^{3/2} - 3x^{5/2} \right). \quad (\text{A4})$$

$$u_3(Q) = \frac{a}{\sqrt{\pi}} \sqrt{6} \left(3x^{3/2} - 12x^{5/2} + 10x^{7/2} \right). \quad (\text{A5})$$

$$u_4(Q) = \frac{a}{\sqrt{\pi}} 2\sqrt{2} \left(4x^{3/2} - 30x^{5/2} + 60x^{7/2} - 35x^{9/2} \right). \quad (\text{A6})$$

$$u_5(Q) = \frac{a}{\sqrt{\pi}} \sqrt{10} \left(5x^{3/2} - 60x^{5/2} + 210x^{7/2} - 280x^{9/2} + 126x^{11/2} \right). \quad (\text{A7})$$

$$u_6(Q) = \frac{a}{\sqrt{\pi}} \sqrt{12} \left(6x^{3/2} - 105x^{5/2} + 560x^{7/2} - 1260x^{9/2} + 1260x^{11/2} - 462x^{13/2} \right). \quad (\text{A8})$$

$$u_7(Q) = \frac{a}{\sqrt{\pi}} \sqrt{14} \left(7x^{3/2} - 168x^{5/2} + 1260x^{7/2} - 4200x^{9/2} + 6930x^{11/2} - 5544x^{13/2} + 1716x^{15/2} \right). \quad (\text{A9})$$

$$u_8(Q) = \frac{a}{\sqrt{\pi}} 4 \left(8x^{3/2} - 252x^{5/2} + 2520x^{7/2} - 11550x^{9/2} + 27720x^{11/2} - 36036x^{13/2} + 24024x^{15/2} - 6435x^{17/2} \right). \quad (\text{A10})$$

From Eq. (A2), we expand the shifted Jacobi polynomial in monomial form,

$$P_{n-1}^{(1,0)}(1-2x) = \sum_{j=0}^{n-1} \hat{c}_{n,j} x^j, \quad (\text{A11})$$

so that the basis function becomes

$$u_n(Q) = \frac{a}{\sqrt{\pi}} \sqrt{2n} \sum_{j=0}^{n-1} \hat{c}_{n,j} x(Q)^{3/2+j}. \quad (\text{A12})$$

The coefficient vectors $\{\hat{c}_{n,j}\}_{j=0}^{n-1}$ are the monomial coefficients of $P_{n-1}^{(1,0)}(1-2x)$:

$$\begin{aligned} \{\hat{c}_{1,j}\} &= \{1\}, \\ \{\hat{c}_{2,j}\} &= \{2, -3\}, \\ \{\hat{c}_{3,j}\} &= \{3, -12, 10\}, \\ \{\hat{c}_{4,j}\} &= \{4, -30, 60, -35\}, \\ \{\hat{c}_{5,j}\} &= \{5, -60, 210, -280, 126\}, \\ \{\hat{c}_{6,j}\} &= \{6, -105, 560, -1260, 1260, -462\}, \\ \{\hat{c}_{7,j}\} &= \{7, -168, 1260, -4200, 6930, -5544, 1716\}, \\ \{\hat{c}_{8,j}\} &= \{8, -252, 2520, -11550, 27720, -36036, 24024, -6435\}. \end{aligned}$$

Using Eq. (39), the Hankel transform of a single term is

$$\tilde{\psi}_{3/2+j}(k) = \frac{1}{2^{j+1/2} a^2 \Gamma(j + \frac{3}{2})} \left(\frac{k}{a}\right)^{j+1/2} K_{j+1/2}\left(\frac{k}{a}\right). \quad (\text{A13})$$

Therefore, applying the transform term by term gives

$$\tilde{u}_n(k) = \frac{a}{\sqrt{\pi}} \sqrt{2n} \sum_{j=0}^{n-1} \hat{c}_{n,j} \tilde{\psi}_{3/2+j}(k), \quad (\text{A14})$$

i.e.,

$$\tilde{u}_n(k) = \frac{\sqrt{2n}}{a\sqrt{\pi}} \sum_{j=0}^{n-1} \frac{\hat{c}_{n,j}}{2^{j+1/2} \Gamma(j + \frac{3}{2})} B^{j+1/2} K_{j+1/2}(B), \quad (\text{A15})$$

where $B = k/a$ is the reduced transform variable.

Using the identity

$$\Gamma(j + \frac{3}{2}) = \frac{(2j+1)!!}{2^{j+1}} \sqrt{\pi}, \quad (\text{A16})$$

this simplifies to

$$\tilde{u}_n(k) = \frac{2\sqrt{n}}{a\pi} \sum_{j=0}^{n-1} \frac{\hat{c}_{n,j}}{(2j+1)!!} B^{j+1/2} K_{j+1/2}(B), \quad (\text{A17})$$

where $(2j+1)!! = 1 \cdot 3 \cdot 5 \cdots (2j+1)$.

Explicitly, the first eight transforms are:

$$\tilde{u}_1(k) = \frac{2}{a\pi} \left(B^{1/2} K_{1/2}(B) \right), \quad (\text{A18})$$

$$\tilde{u}_2(k) = \frac{2\sqrt{2}}{a\pi} \left(2B^{1/2} K_{1/2}(B) - B^{3/2} K_{3/2}(B) \right), \quad (\text{A19})$$

$$\begin{aligned} \tilde{u}_3(k) = \frac{2\sqrt{3}}{a\pi} & \left(3B^{1/2} K_{1/2}(B) - 4B^{3/2} K_{3/2}(B) \right. \\ & \left. + \frac{2}{3} B^{5/2} K_{5/2}(B) \right), \end{aligned} \quad (\text{A20})$$

$$\begin{aligned} \tilde{u}_4(k) = \frac{4}{a\pi} & \left(4B^{1/2} K_{1/2}(B) - 10B^{3/2} K_{3/2}(B) \right. \\ & \left. + 4B^{5/2} K_{5/2}(B) - \frac{1}{3} B^{7/2} K_{7/2}(B) \right), \end{aligned} \quad (\text{A21})$$

$$\begin{aligned} \tilde{u}_5(k) = \frac{2\sqrt{5}}{a\pi} & \left(5B^{1/2} K_{1/2}(B) - 20B^{3/2} K_{3/2}(B) \right. \\ & + 14B^{5/2} K_{5/2}(B) - \frac{8}{3} B^{7/2} K_{7/2}(B) \\ & \left. + \frac{2}{15} B^{9/2} K_{9/2}(B) \right), \end{aligned} \quad (\text{A22})$$

$$\begin{aligned} \tilde{u}_6(k) = \frac{2\sqrt{6}}{a\pi} & \left(6B^{1/2} K_{1/2}(B) - 35B^{3/2} K_{3/2}(B) \right. \\ & + \frac{112}{3} B^{5/2} K_{5/2}(B) - 12B^{7/2} K_{7/2}(B) \\ & \left. + \frac{4}{3} B^{9/2} K_{9/2}(B) - \frac{2}{45} B^{11/2} K_{11/2}(B) \right), \end{aligned} \quad (\text{A23})$$

$$\begin{aligned} \tilde{u}_7(k) = \frac{2\sqrt{7}}{a\pi} & \left(7B^{1/2} K_{1/2}(B) - 56B^{3/2} K_{3/2}(B) \right. \\ & + 84B^{5/2} K_{5/2}(B) - 40B^{7/2} K_{7/2}(B) \\ & + \frac{22}{3} B^{9/2} K_{9/2}(B) - \frac{8}{15} B^{11/2} K_{11/2}(B) \\ & \left. + \frac{4}{315} B^{13/2} K_{13/2}(B) \right), \end{aligned} \quad (\text{A24})$$

$$\begin{aligned} \tilde{u}_8(k) = \frac{4\sqrt{2}}{a\pi} & \left(8B^{1/2} K_{1/2}(B) - 84B^{3/2} K_{3/2}(B) \right. \\ & + 168B^{5/2} K_{5/2}(B) - 110B^{7/2} K_{7/2}(B) \\ & + \frac{88}{3} B^{9/2} K_{9/2}(B) - \frac{52}{15} B^{11/2} K_{11/2}(B) \\ & + \frac{8}{45} B^{13/2} K_{13/2}(B) - \frac{1}{315} B^{15/2} K_{15/2}(B) \left. \right). \end{aligned} \quad (\text{A25})$$

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²⁰The argument $1 - 2x$ arises from the affine mapping $t = 1 - 2x$, which transforms the standard Jacobi orthogonality interval $t \in [-1, 1]$ to the compact scattering variable $x \in [0, 1]$. This transformation introduces a Jacobian factor $|dt/dx| = 2$. Specifically, substituting $t = 1 - 2x$ into the orthogonality relation $\int_{-1}^1 (1-t)^\alpha P_n^{(\alpha,0)}(t) P_m^{(\alpha,0)}(t) dt = h_n \delta_{nm}$ yields

$$\int_0^1 (2x)^\alpha P_n^{(\alpha,0)}(1-2x) P_m^{(\alpha,0)}(1-2x) (2dx) = h_n \delta_{nm}.$$

Extracting the constant factors $2^{\alpha+1}$ and dividing by the norm $h_n = \frac{2^{\alpha+1}}{2n+\alpha+1}$ (with $\alpha = 2b-2$) recovers the normalization condition $\int_0^1 x^{2b-2} [P_n]^2 dx = (2n+2b-1)^{-1}$.

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